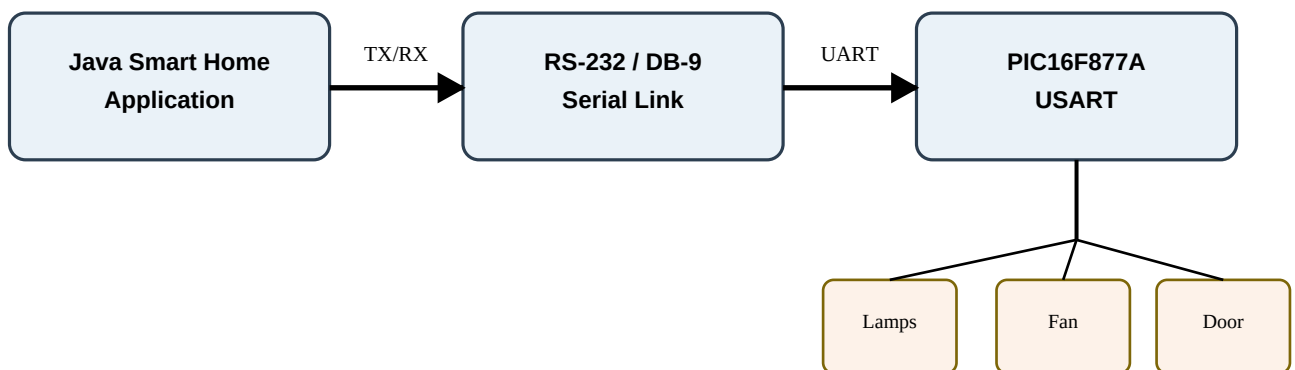


PC-BASED SMART HOME AUTOMATION

Serial Communication Parameters & Command Protocol

Technical documentation for the Java desktop application, RS-232 serial interface, and PIC16F877A embedded controller.

Bidirectional Serial Communication Architecture



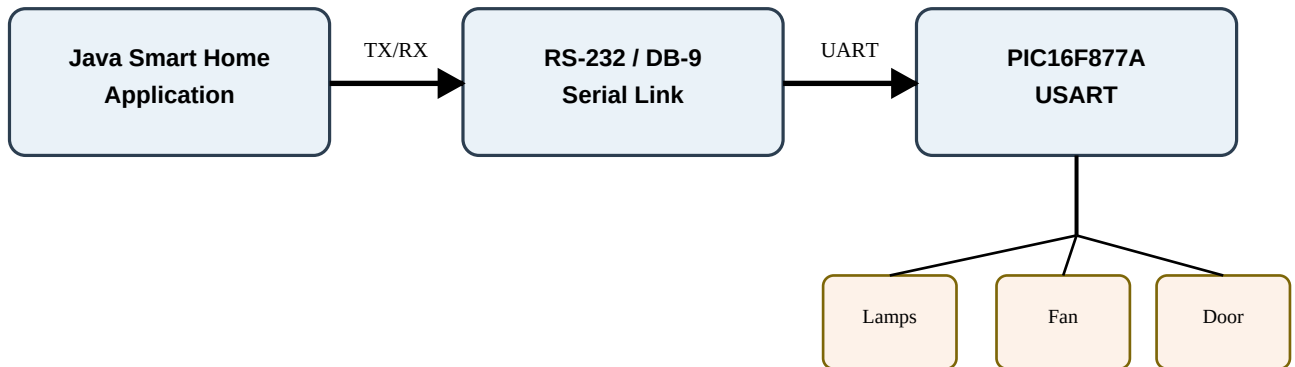
Project scope: Bidirectional communication is used to control home appliances from a PC application and receive system information such as appliance status and temperature.

Repository directory: [serial-communication/](#)

1. System Overview

The system combines a Java desktop GUI with a PIC16F877A embedded controller. The PC sends control commands through a serial COM port. The microcontroller interprets the received characters and drives the corresponding appliance-control outputs. Feedback can be returned to the PC for monitoring.

Bidirectional Serial Communication Architecture



2. Serial Communication Parameters

Parameter	Value
Communication interface	RS-232 serial communication
PC interface	COM port / DB-9 serial interface
Microcontroller	PIC16F877A
UART	Hardware USART
Baud rate	9600 bps
Data bits	8
Parity	None
Stop bits	1
Flow control	None
Frame format	8-N-1
Communication mode	Bidirectional
PC application	Java desktop application (Swing)
Serial library	jSerialComm
Firmware development	MikroC PRO for PIC

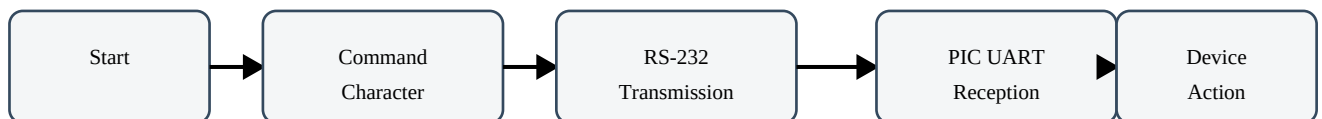
3. Character Command Protocol

The following table documents the single-character control scheme used for the appliance-control concept. The exact characters should match the final Java and MikroC source files before release.

Command	Function	Direction
A	Turn ON Lamp 1	PC → PIC
a	Turn OFF Lamp 1	PC → PIC
B	Turn ON Lamp 2	PC → PIC
b	Turn OFF Lamp 2	PC → PIC
C	Turn ON Lamp 3	PC → PIC
c	Turn OFF Lamp 3	PC → PIC
F	Turn ON Fan	PC → PIC
f	Turn OFF Fan	PC → PIC
D	Open / activate Door Motor	PC → PIC
d	Close / deactivate Door Motor	PC → PIC

4. Command Transmission Flow

Typical PC → Microcontroller Command Path



Example: When the user selects Lamp 1 ON in the Java application, the application transmits the character **A**. The PIC16F877A receives the character through its USART and activates the corresponding output.

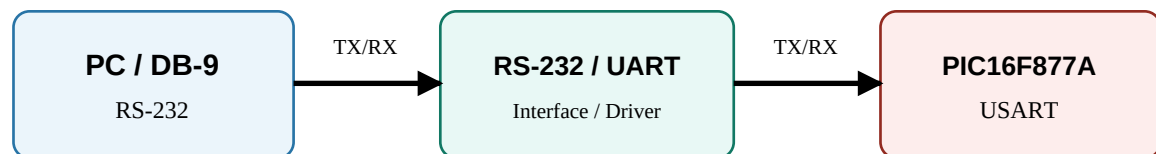
5. Feedback and Monitoring

The PIC16F877A can transmit information back to the Java application over the serial connection. The Java GUI can use this data to update appliance states, display temperature information, and maintain a system log.

Feedback category	Purpose
Appliance status	Display current ON/OFF state in the Java GUI
Temperature	Update the temperature display
Door status	Indicate the current door state
System messages	Provide communication or operating-status information

6. RS-232 / UART Interface Consideration

Note: RS-232 electrical levels require the appropriate interface/driver; do not connect RS-232 voltage levels directly to PIC TTL/CMOS UART pins.



The PC serial interface and PIC UART use different electrical signaling conventions depending on the interface hardware. An appropriate RS-232 level converter/driver must be used where required. The documentation should identify the actual interface circuit used in the Proteus design and hardware prototype.

7. Recommended Repository Contents

File / Folder	Purpose
README.md	Short project overview and links to documentation
serial-communication.pdf	This technical communication document
java/	Java desktop application source code
mikroc/	PIC16F877A firmware source code
proteus/	Serial interface and system simulation files
images/	Circuit, GUI, and hardware photographs
videos/	Optional demonstration videos or links

Document note: The command characters shown above are the proposed/previously discussed command mapping. Before treating this PDF as the final project specification, verify the characters against the latest Java and MikroC source code.